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(54) **SYSTEM AND METHOD CONFIGURED TO GENERATE A MEDIA PRESENTATION USING GENERATIVE ARTIFICIAL INTELLIGENCE**

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G06T 11/60 (2006.01)

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(58) **Field of Classification Search**
CPC G06T 11/60; G06F 40/279; G06F 40/40
See application file for complete search history.

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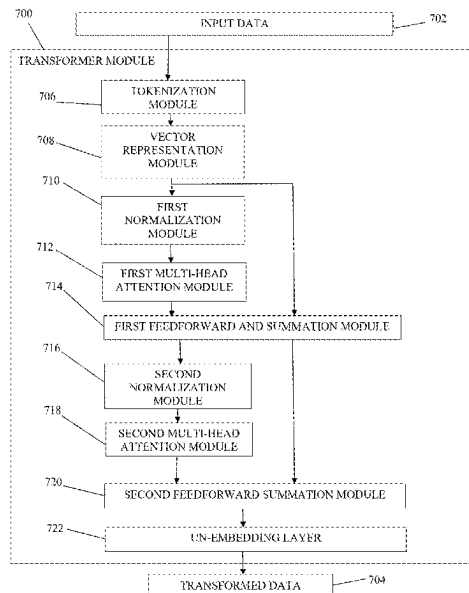
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(57) **ABSTRACT**

A system and method are configured to generate an enhanced media presentation from an initial media presentation having initial text using generative artificial intelligence (AI). An extraction module include a first generative AI module to extract a plurality of terms from the initial text. A prompt generating module includes a second generative AI module to generate a prompt from the plurality of terms. A presentation generating module includes a third generative AI module to generate enhancement media from the prompt, and to generate and output an enhanced media presentation combining the initial text with the enhancement media. The method implements the system.

17 Claims, 13 Drawing Sheets



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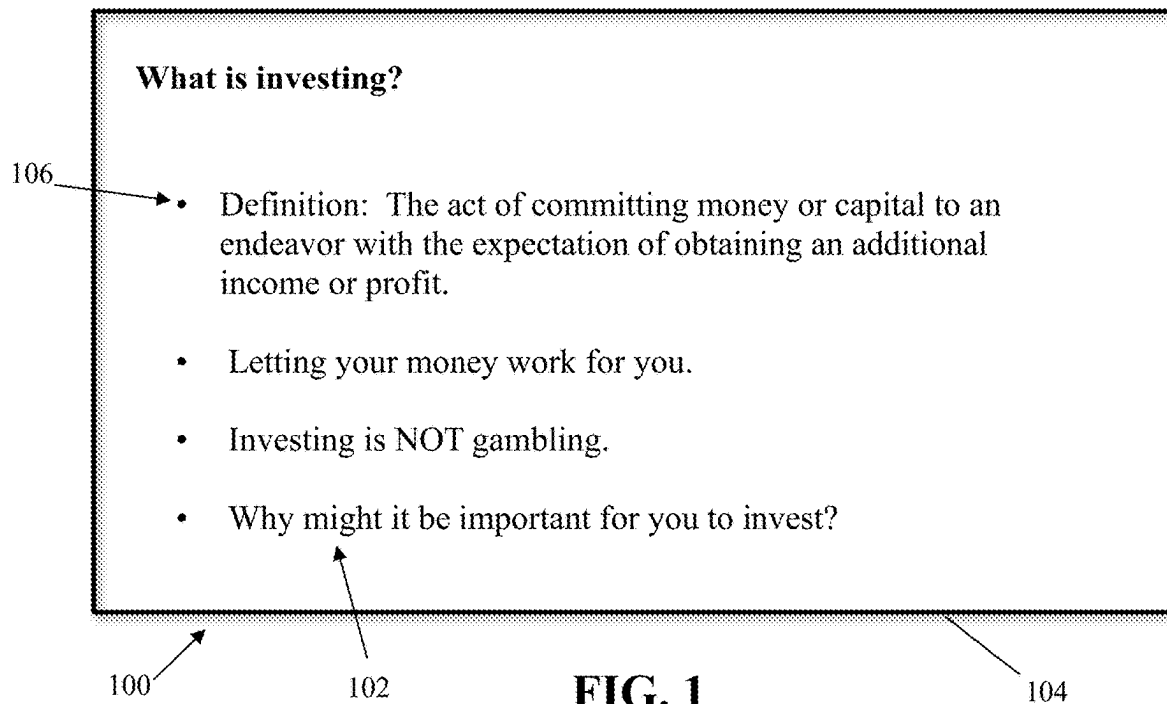
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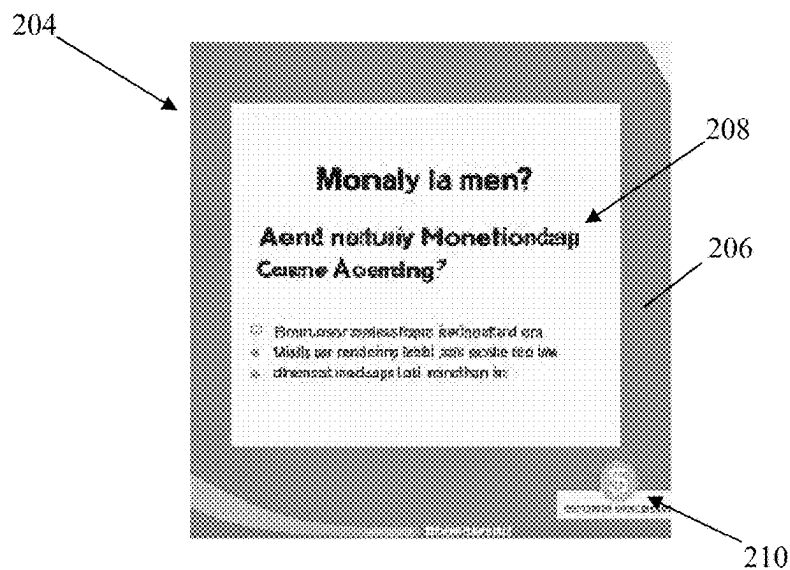
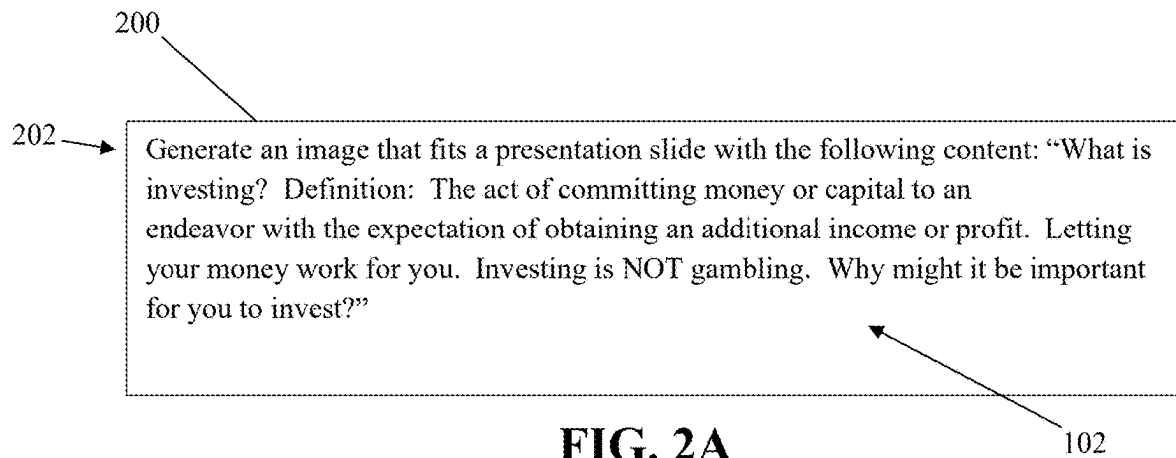
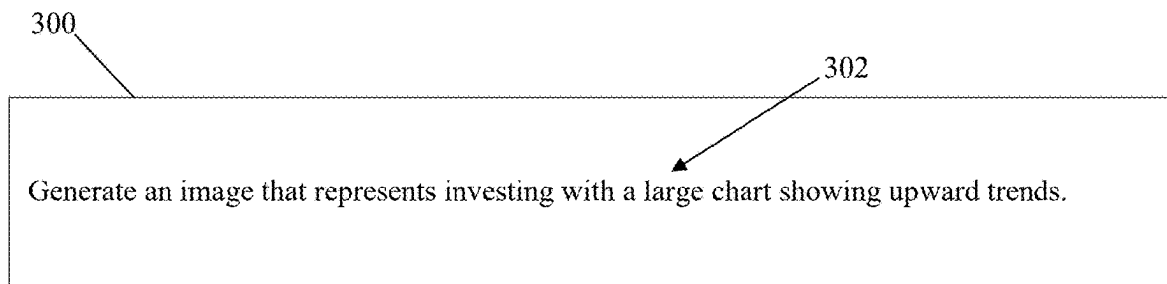
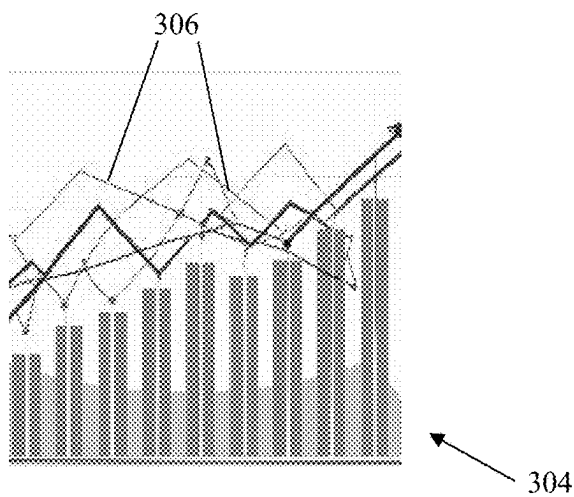


FIG. 2B
(PRIOR ART)

**FIG. 3A****FIG. 3B**
(PRIOR ART)

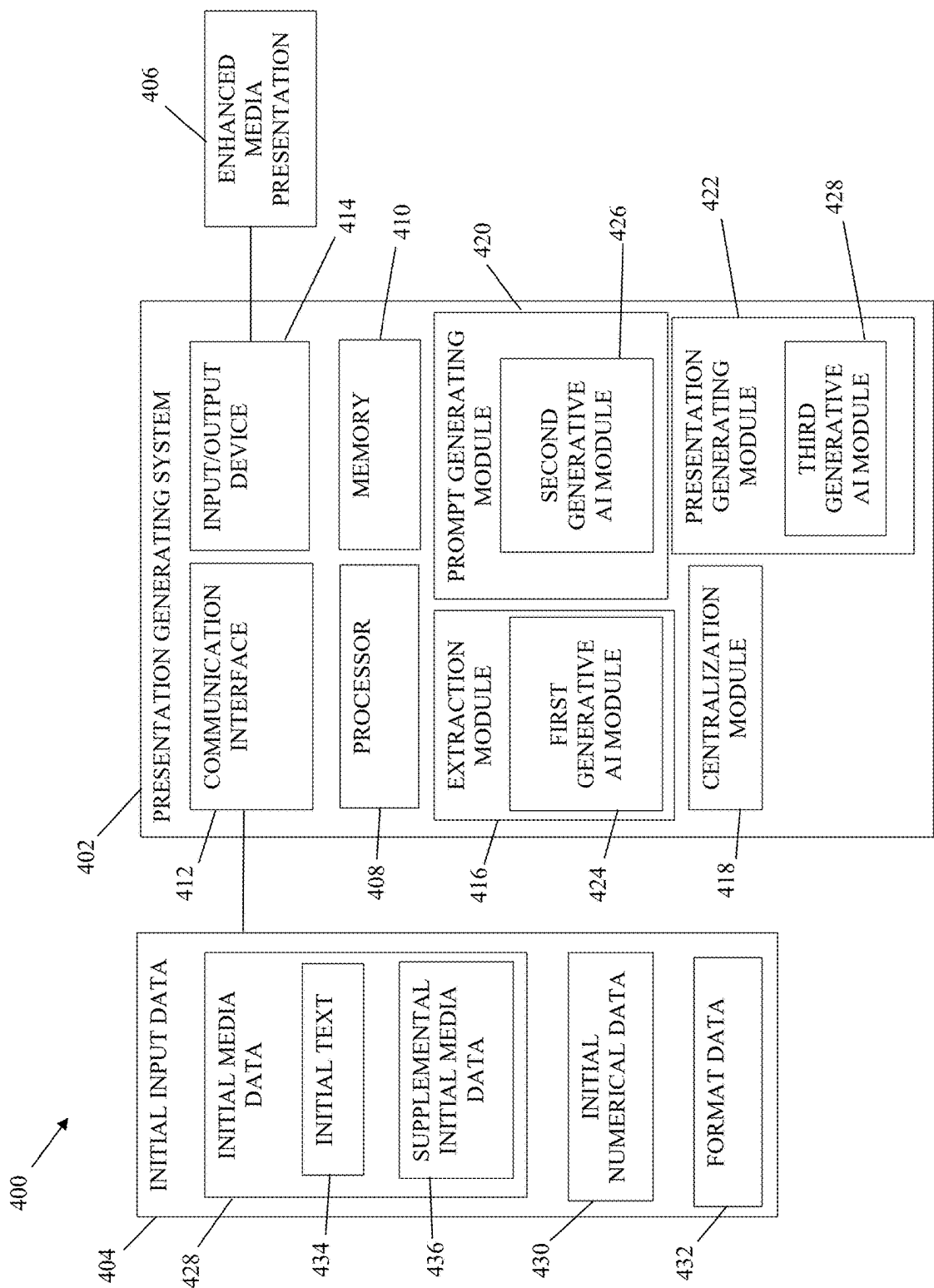
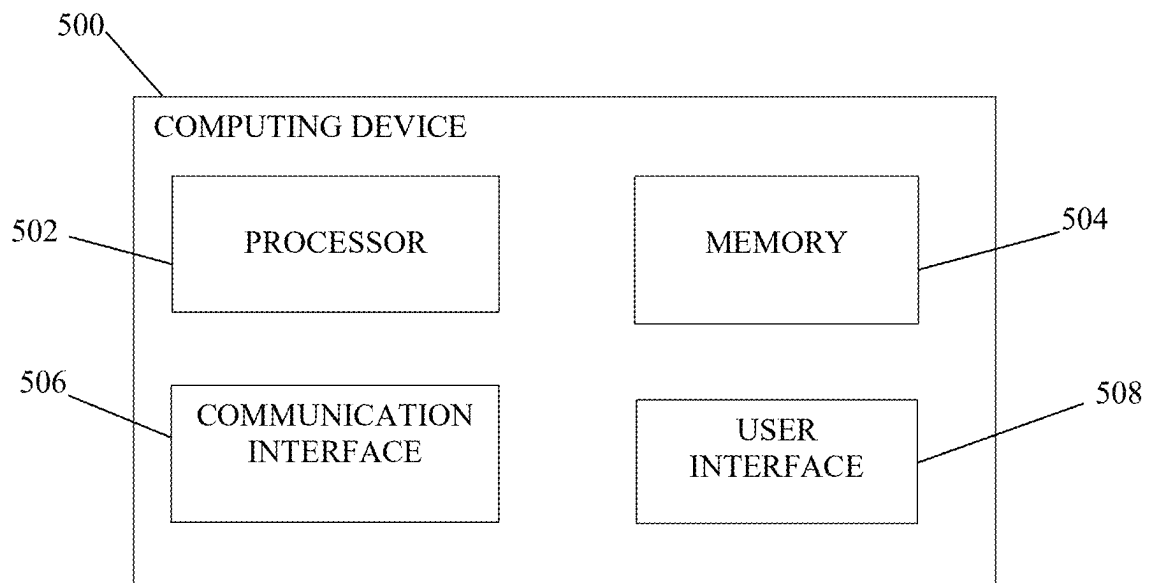


FIG. 4

**FIG. 5**

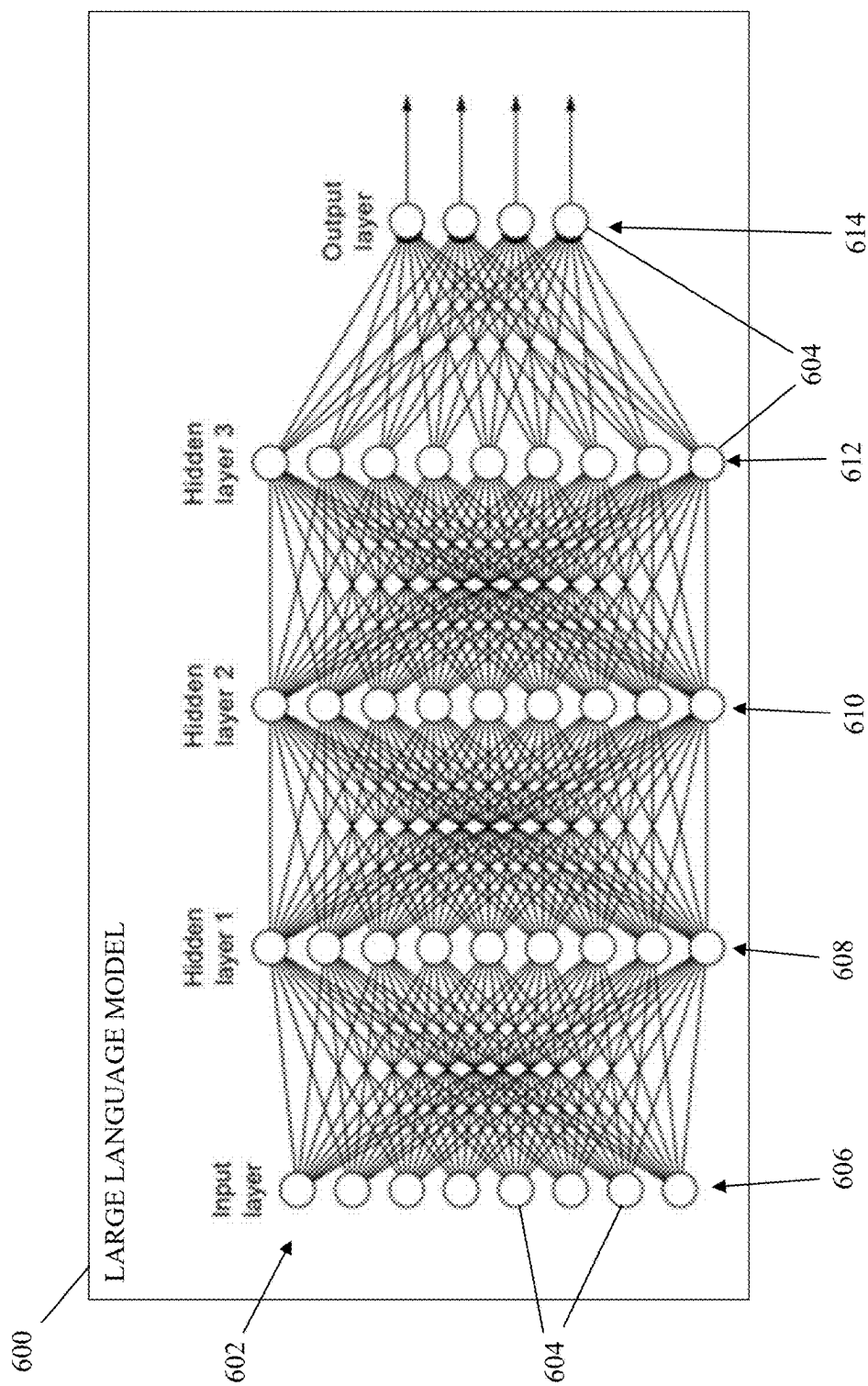


FIG. 6

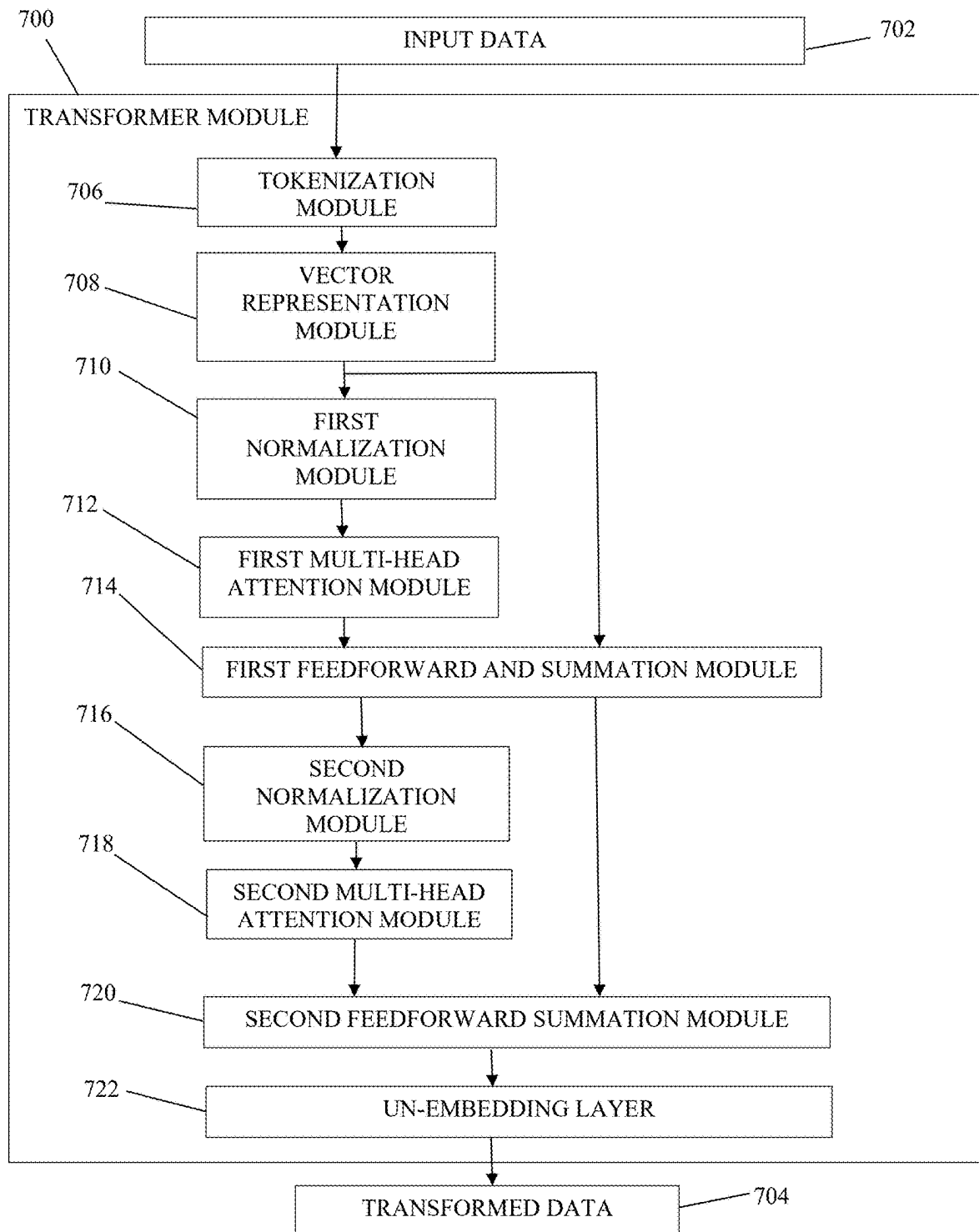


FIG. 7

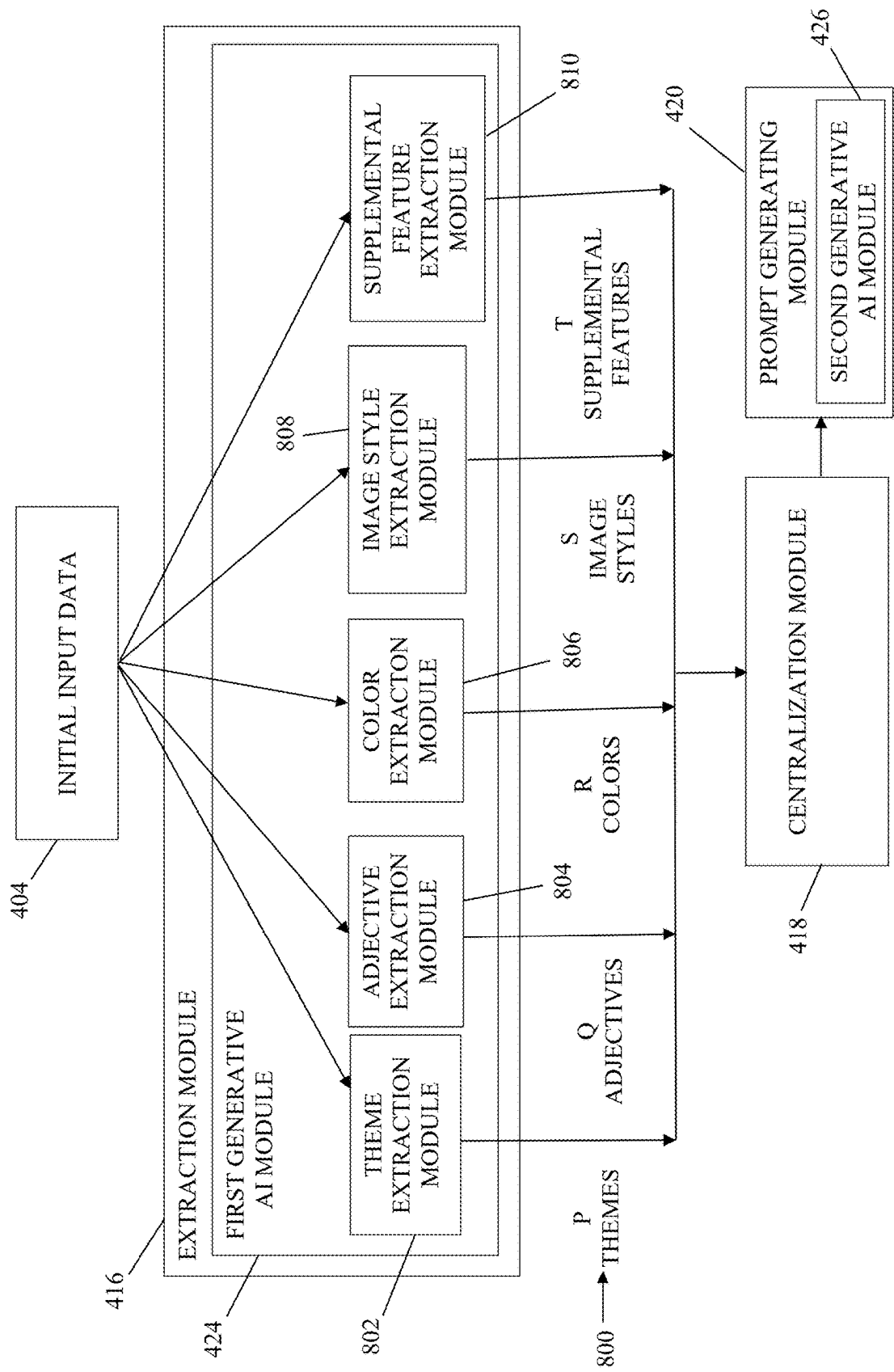


FIG. 8

900	902	904	906
EXTRACTED FEATURE	DEFAULT NUMBER OF TERMS	INSTRUCTIONS TO THE FIRST GENERATIVE AI MODULE	
Theme	P=1	From the contents of the input slide, extract P terms that represents the theme of the presentation in a very generic way. The terms selected should abstract away from the specifics of the input slide. Only list the word in your answer. Examples are law, technology, art, science, and business.	
Adjective	Q=5	From the contents of the input slide, extract Q terms that represent important adjectives that describe the contents. The adjectives should not paraphrase the content. The adjectives selected should be relevant to the text from the input slide.	
Color	R=5	From the contents of the input slide, extract R terms as comma-separated names of colors that fit the ambiance of the input slide.	
Image Style	S=5	From the contents of the input slide, extract S terms as comma-separated terms that describing possible art and image styles that could fit in an accompanying image.	

FIG. 9

1000	1002	1004
EXTRACTED FEATURE	TERMS DESCRIBING THE FEATURE	
Theme	Investment	-
Adjective	Committing	Important
Color	Green	Grey
Image Style	Financial	Professional
		Investing
		White
		Simplistic

FIG. 10

1100

Based on the extracted features and terms of the extracted features, create a prompt for a scene using the extracted features and terms. The prompt is to describe every detail of a scene. The prompt should use at least one theme, at least one adjective, at least one color, and at least one image among the terms provided. The scene should be minimalist and creative. The prompt should be at most 150 words in length.

FIG. 11

1200

In the heart of a *professional* office, a *commitment* to investment is pervasive. A minimalist, *light-grey* table with organized piles of *blue* and *white* reports radiate an air of importance.

On its polished surface, a *green* laptop flickers with an infographic detailing profitable ventures, adorned with *simple* additional yet crucial data.

Golden rays seep through the sleek window, delicately stroking the laptop's keyboard. The room exudes a distinct *educational* ambiance - a silent seminar in *investing*.

Every corner bears witness to the purposeful hustle, the passionate pursuit of *financial* wisdom found through the simplistic interpretation of numbers; a scene that subtly illustrates the heart of *investment*.

FIG. 12

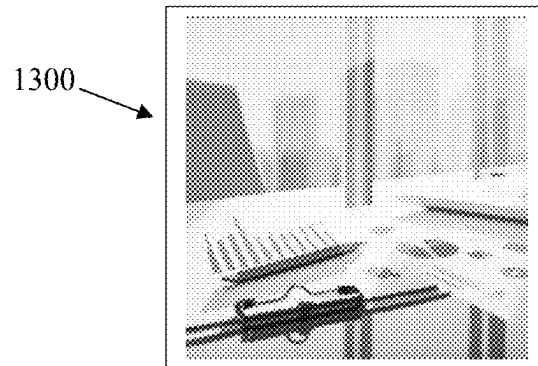


FIG. 13A

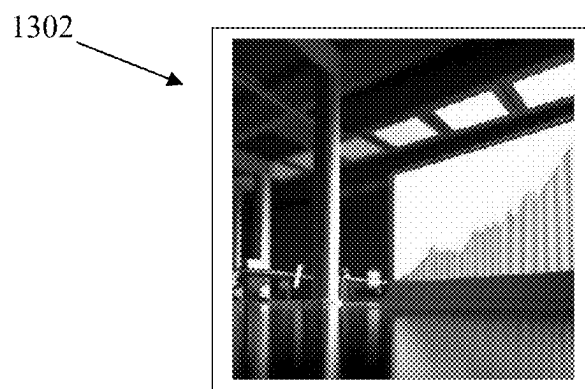


FIG. 13B

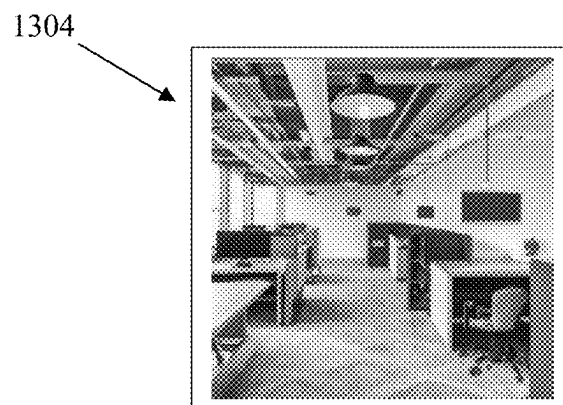
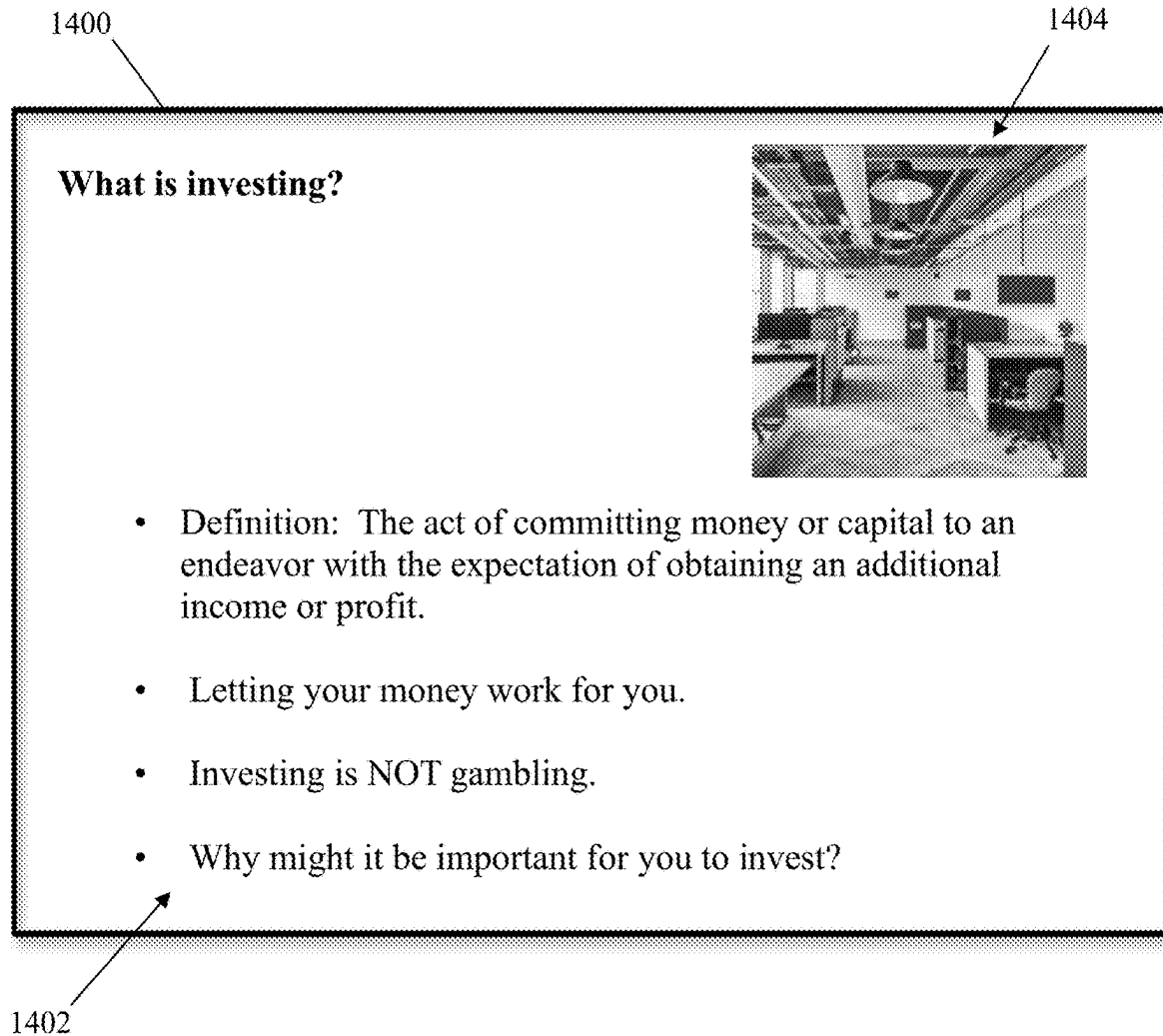
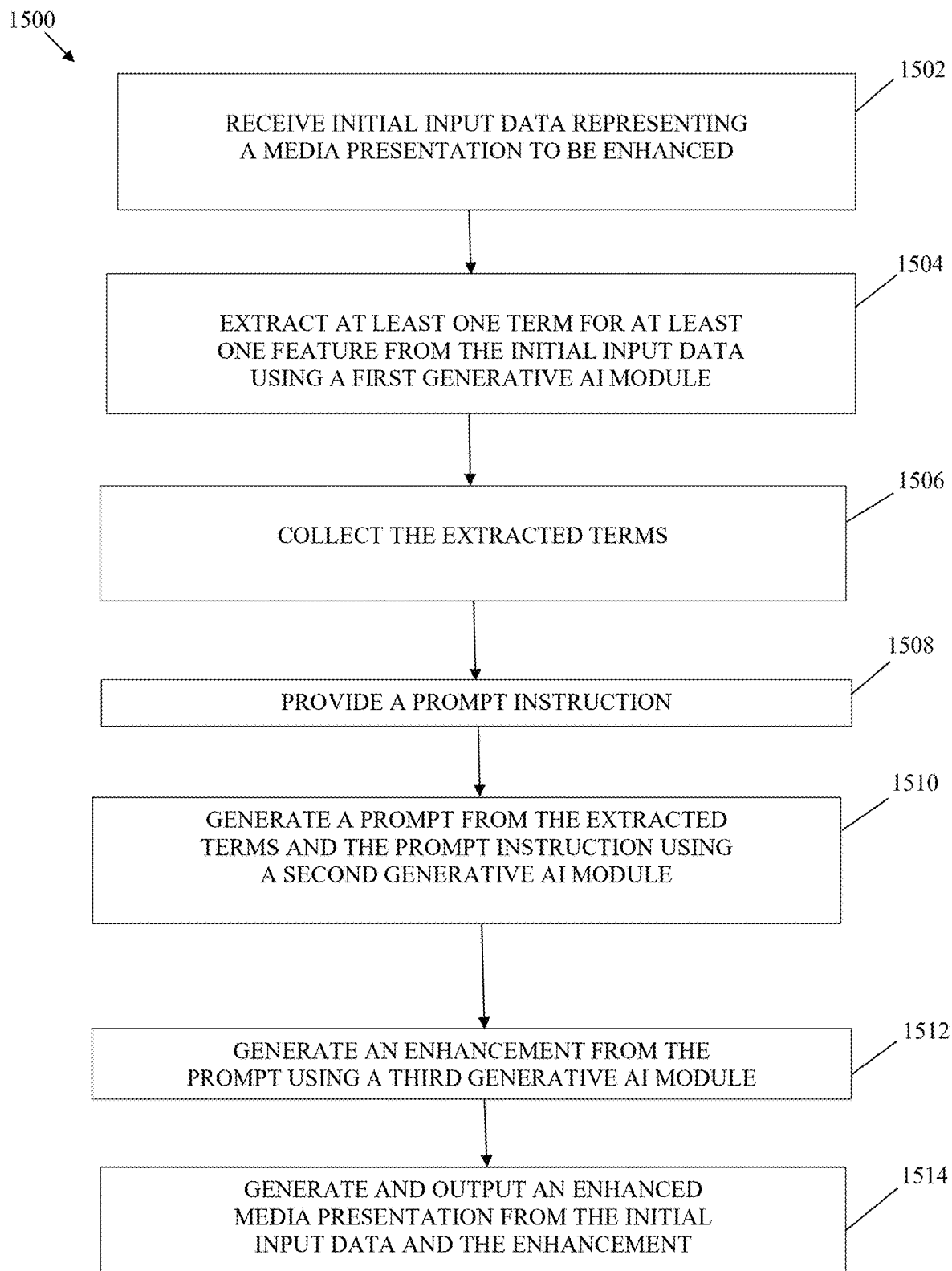


FIG. 13C

**FIG. 14**

**FIG. 15**

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SYSTEM AND METHOD CONFIGURED TO GENERATE A MEDIA PRESENTATION USING GENERATIVE ARTIFICIAL INTELLIGENCE

FIELD OF THE DISCLOSURE

The present disclosure relates generally to media presentations, and, more particularly, to a system and method configured to generate a media presentation using generative artificial intelligence.

BACKGROUND OF THE DISCLOSURE

Media presentations are very important to convey information in a summary format to an audience. For example, media presentations employ visual charts of data summarizing large tables of data, since visual charts convey trends which displayed raw numerical data cannot readily convey in a short time. In addition, text on consecutive slides describing ideas to an audience are often summarized using short phrases as well as bullet points. Good visuals are hard to find, and so media presentations with large numbers of words and numbers are usually ignored by the audience and have the effect of a very dry delivery of ideas by the presenter. Accordingly, despite such use of charts, bullet points, and other features used in media presentations, including cartoons, which are meant to quickly convey information, there is a low engagement rate of audiences to media presentations.

Artificial intelligence (AI) has many applications, including the creation of new media, such as text, an audio file, an image, an animation, and a video. Therefore, the use of AI in creating a media presentation is promising to counter the dry delivery and low engagement rate of known media presentations. However, even though AI has evolved, on occasion, an AI engine may create media having content including gibberish from the human perspective. In the field of media presentations, media is created and output from known presentation software, such as POWERPOINT publicly available from MICROSOFT CORPORATION. However, when AI is applied to presentation software to create media presentations, any generated content including gibberish can embarrass a presenter.

For example, as shown in FIG. 1, a user can generate a slide **100** including text **102**. In one implementation, the user manually generates the slide **100** by inputting text **102** and commands into known presentation software. In addition to the text **102**, the slide **100** optionally includes formatting such as a border **104** around the periphery of the slide **100**, and bullet points **106** corresponding to portions of the text **102**. In one implementation, the slide **100** is a page generated and displayed by presentation software. In another implementation, the slide **100** is a webpage with the text **102**, generated using webpage authoring software. For example, the webpage stores the text **102** in a HyperText Markup Language (HTML). The webpage uses delimiters positioned on each side of a line or multiple lines of text **102**, such as the delimiters “<>” and “</>”. For example, the text **102** includes tags such as “<body>” and “</body>” on either side of the plain text to be displayed on the webpage. The webpage-based slide **100** is displayed by a web browser as presentation software.

Attempts have been made to utilize AI to automatically enhance such a slide **100**. In one implementation, AI has been utilized to add an image to the slide **100** in response to a prompt, such as the example prompt **200** shown in FIG. 2A

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having a command **202** and the text **102** of the slide **100**. In one implementation, a user inputs the command **202** as well as the text **102** to generate the prompt **200** in natural language, such as English or a preset natural language. The prompt **200** is applied to an AI-based computer program, such as the DALL-E model, publicly available from OPENAI, to add an image to the slide **100** or to generate an enhanced slide **204**, as shown in FIG. 2B. Such AI-based computer programs can create realistic images and art from a description in natural language.

For example, using an AI-based computer program, the enhanced slide **204** includes a border **206** around the text **208**, as well as other images **210**. In addition, the text **208** is derived from the text **102** in the slide **100** shown in FIG. 1. However, known AI-based computer programs, such as DALL-E, occasionally generate such an enhanced slide **204** with the text **208** garbled to appear as gibberish to a human viewer, as shown in FIG. 2B. In addition, the known AI-based computer programs garble the images **206**, **210**. For example, in FIG. 2B, the image **210** appears to be a distorted dollar sign “\$”.

In another implementation shown in FIGS. 3A-3B, a user generates a prompt **300** with a command **302** to generate an image to be associated with the slide **100** in FIG. 1. In the example shown in FIG. 3A, the prompt **300** instructs an AI-based computer program, such as DALL-E, to generate an image that represents investing with a large chart showing upward trends. The prompt **300** is applied to an AI-based computer program, such as DALL-E, to generate such an image **304**, as shown in FIG. 3B. However, known AI-based computer programs, such as DALL-E, occasionally generate inaccurate, garbled, or inappropriate images, such as the image **304** which illustrates a chart with numerous plot lines **306** going in multiple directions instead of the desired upward or a generally upward trend.

Such images **204**, **304** in FIGS. 2B and 3B, respectively, generated by known AI-based computer programs, fail to serve the intended purpose of applying AI to presentations; that is, to present and enhance information to convey additional information and to improve engagement of the viewer with the content of the presentation.

Other known problems with AI-based computer programs used to generate media presentations include the time required to write a sufficiently detailed or lengthy description for each slide in a presentation, the occurrence of meaningless combinations of letters and numbers, and repetition of images or similar images when a single topic is being summarized.

SUMMARY OF THE DISCLOSURE

According to an implementation consistent with the present disclosure, a system and method are configured to generate a media presentation using generative artificial intelligence.

In an implementation, a presentation enhancement system is configured to enhance an initial media presentation having initial text. The presentation enhancement system includes a hardware-based processor, a memory, and a set of modules. The memory configured to store instructions and configured to provide the instructions to the hardware-based processor. The set of modules are configured to implement the instructions provided to the hardware-based processor. The set of modules includes an extraction module, a prompt generating module, and a presentation generating module. The extraction module includes a first generative artificial intelligence (AI) module configured to automatically extract a plurality

of terms from the initial text. The prompt generating module includes a second generative AI module configured to automatically generate a prompt from the plurality of terms. The presentation generating module includes a third generative AI module configured to automatically generate enhancement media from the prompt, and to automatically generate and output an enhanced media presentation combining the initial text with the enhancement media.

The third generative AI module can automatically generate an image as the enhancement media from the prompt. The first and second generative AI modules can be text-to-text generative AI modules, wherein the plurality of features and the prompt can be in a text format, and the third generative AI module can be a text-to-image generative AI module configured to automatically generate an image as the enhancement media from the text-formatted prompt. Each of the first, second, and third generative AI modules can include a neural network configured as a transformer to implement a large language model. The plurality of features can include a theme, an adjective, a color, and an image style. The extraction module further comprises a theme extraction module configured to extract the theme from the initial text, an adjective extraction module configured to extract the adjective from the initial text, a color extraction module configured to extract the color from the initial text, and an image style extraction module configured to extract the image style from the initial text.

The third generative AI module can automatically generate the enhancement media from the prompt using a temperature parameter. The value of the temperature parameter can be set to a predetermined temperature value between 0.5 and 1.0, inclusive. Alternatively, the value of the temperature parameter can be set to a randomized temperature value between 0.5 and 1.0, inclusive.

In another implementation, a presentation enhancement system is configured to enhance an initial media presentation having initial text. The presentation enhancement system comprises a hardware-based processor, a memory, and a set of modules. The memory is configured to store instructions and configured to provide the instructions to the hardware-based processor. The set of modules is configured to implement the instructions provided to the hardware-based processor. The set of modules includes an extraction module, a prompt generating module, and a presentation generating module. The extraction module can include a first generative artificial intelligence (AI) module implementing a first large language model and configured to automatically extract a plurality of terms from the initial text. The prompt generating module can include a second generative AI module implementing a second large language model and configured to automatically generate a prompt from the plurality of terms. The presentation generating module can include a third generative AI module implementing a third large language model and configured to automatically generate enhancement media from the prompt, and to automatically generate and output an enhanced media presentation combining the initial text with the enhancement media.

The third generative AI module can automatically generate an image as the enhancement media from the prompt. The first and second generative AI modules can be text-to-text generative AI modules, wherein the plurality of features and the prompt can be in a text format. The third generative AI module can be a text-to-image generative AI module configured to automatically generate an image as the enhancement media from the text-formatted prompt. Each of the first, second, and third generative AI modules can include a neural network configured as a transformer to

implement the first, second, and third large language models, respectively. The plurality of features can include a theme, an adjective, a color, and an image style. The extraction module can further comprise a theme extraction module configured to extract the theme from the initial text, an adjective extraction module configured to extract the adjective from the initial text, a color extraction module configured to extract the color from the initial text, and an image style extraction module configured to extract the image style from the initial text.

The third generative AI module can automatically generate the enhancement media from the prompt using a temperature parameter. The value of the temperature parameter can be set to a predetermined temperature value between 0.5 and 1.0, inclusive. Alternatively, the value of the temperature parameter can be set to a randomized temperature value between 0.5 and 1.0, inclusive.

In a further implementation, a computer-based method comprises receiving initial input data representing an initial media presentation, automatically extracting at least one term for at least one feature from the initial input data using a first generative artificial intelligence (AI) module, providing a prompt instruction, automatically generating a prompt from the extracted at least one term and from the prompt instruction using a second generative AI module, automatically generating an enhancement media from the prompt using a third generative AI module, and generating and outputting an enhanced media presentation from the initial input data and the enhancement media. Each of the first, second, and third generative AI modules can include a neural network configured as a transformer to implement a large language model.

Any combinations of the various embodiments, implementations, and examples disclosed herein can be used in a further implementation, consistent with the disclosure. These and other aspects and features can be appreciated from the following description of certain implementations presented herein in accordance with the disclosure and the accompanying drawings and claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example of a presentation slide to be enhanced.

FIG. 2A is a prompt to generate an image for the slide of FIG. 1.

FIG. 2B illustrates an example of an AI-generated image using the prompt in FIG. 2A.

FIG. 3A is a prompt to generate an image representing a chart.

FIG. 3B illustrates an example of an AI-generated chart using the prompt in FIG. 3A.

FIG. 4 is a schematic of a system, according to an implementation.

FIG. 5 is a schematic of a computing device used in the implementation.

FIG. 6 is a schematic of a large language model including a neural network.

FIG. 7 is a schematic of a transformer module used in the large language model of FIG. 6.

FIG. 8 is a schematic of the extraction process employed by the system of FIG. 4.

FIG. 9 is a table of instructions to perform extraction of terms.

FIG. 10 is a table of extraction results.

FIG. 11 is an instruction to create a prompt based on the extraction results of FIG. 10.

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FIG. 12 is a prompt generated using the instruction of FIG. 11.

FIGS. 13A-13C are different images generated from the prompt in FIG. 12.

FIG. 14 is an enhanced media presentation.

FIG. 15 is a flowchart of operation of the system of FIG. 4.

It is noted that the drawings are illustrative and are not necessarily to scale.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS OF THE DISCLOSURE

Example embodiments and implementations consistent with the teachings included in the present disclosure are directed to a system 400 and method 1500 configured to generate a media presentation using generative artificial intelligence (AI).

Referring to FIG. 4, in an implementation consistent with the invention, the system 400 includes a presentation generating system 402 configured to receive initial input data 404 and to automatically generate an enhanced media presentation 406. The generated enhanced media presentation 406 is a presentation which enhances the initial input data 404 with enhancement generated using generative AI techniques. In an implementation, the initial input data 404 specifies an initial media presentation including media, such as text as well as an image, audio, animation, video, numerical data, a graph representing numerical data, or any other known media. For example, the initial input data 404 specifies a slide for use in presentations, such as the slide 100 shown in FIG. 1, as described below in greater detail. The generated enhanced media presentation 406 is an enhanced version of the initial media presentation such as the slide 100. For example, the generated enhanced media presentation 406 includes at least one image added to the text 102 in FIG. 1, such that the added at least one image complements, supplements, or is otherwise relevant to the text 102 or other information presented on the slide 100. In another example, the generated enhanced media presentation 406 includes at least one of text, an image, audio, animation, video, numerical data, a graph representing numerical data, or any other known media configured to complement, supplement, or otherwise to be relevant to the text 102 or other information presented on the slide 100.

In one implementation, the presentation generating system 402 includes a hardware-based processor 408, a memory 410 configured to store instructions and configured to provide the instructions to the hardware-based processor 408, a communication interface 412, an input/output device 414, and a set of modules 416-428 configured to implement the instructions provided to the hardware-based processor 408. In an implementation, the set of modules 416-428 include an extraction module 416, a centralization module 418, a prompt generating module 420, and a presentation generating module 422. The extraction module 416 includes a first generative AI module 424 configured to perform extraction operations using generative AI techniques, as described below. The prompt generating module 420 includes a second generative AI module 426 configured to perform prompt generating operations using generative AI techniques, as described below. The presentation generating module 422 includes a third generative AI module 428 configured to generate a presentation using generative AI techniques, as described below. In another implementation, the presentation generating system 402 operates the set of modules 416-428 using an application written in the

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PYTHON programming language. In a further implementation, the presentation generating system 402 operates the set of modules 416-428 using an application written in FLASK, which is a small and lightweight PYTHON software framework. In still another implementation, the presentation generating system 402 operates the set of modules 416-428 using any known operating system or control application written in any known programming language.

In one implementation, the first generative AI module 424, the second generative AI module 426, and the third generative AI module 428 are separate and independent from each other. In an alternative implementation, the first generative AI module 424, the second generative AI module 426, and the third generative AI module 428 are integrated into a single generative AI module configured to perform generative AI operations for the extraction module 416, for the prompt generating module 420, and for the presentation generating module 422, as described below. In another alternative implementation, since the third generative AI module 428 is a text-to-image generating module, as described in greater detail below, while the first generative AI module 424 and the second generative AI module 426 are text-to-text generating modules, as described in greater detail below, the first generative AI module 424 and the second generative AI module 426 are integrated into a single generative AI module configured to perform generative AI operations for the extraction module 416 and for the prompt generating module 420, as described below.

In one implementation, the system 400 is operatively connected to a data source through a network to transmit, convey, or otherwise provide the initial input data 404 to the presentation generating system 402. For example, the network is the Internet. In another example, the network is an internal network or intranet of an organization. In a further example, the network is a heterogeneous or hybrid network including the Internet and the intranet. In one implementation, the data source is a database configured to store the initial input data 404. In another implementation, the data source is a user device through which a user generates the initial input data 404, for example, by user inputs, selections, or commands from the user to the user device.

In an implementation consistent with the invention, the initial input data 404 includes initial media data 428. Optionally, the initial input data 404 includes initial numerical data 430 and format data 432. The initial media data 428 includes initial text 434 and optionally supplemental initial media data 436. For example, the initial input data 404 specifies the slide 100 including the text 102 shown in FIG. 1. In an implementation, the initial text 434 is in a text-based format. In one implementation, the slide 100 is a page generated and displayed by known presentation software, such as POWERPOINT publicly available from MICROSOFT CORPORATION. In an implementation, the user manually generates the slide 100 by inputting the text 102 as the initial text 434 as well as commands or other selections into the known presentation software. In one implementation, the text 102 is a text file having the American Standard Code for Information Interchange (ASCII) file format. In addition to the text 102, the slide 100 optionally includes formatting such as a border 104 around the periphery of the slide 100, and bullet points 106 corresponding to portions of the text 102. For example, the formatting for the border 104 and the bullet points 106 are specified by the format data 432.

In another implementation, the slide 102 includes other media, such as audio files, video files, animation files, graphical image data, or other known media specified by the supplemental initial media data 436. For example, the audio

files are Waveform Audio File Format (WAV) files, the video files are Moving Picture Experts Group (MPEG) files, and the animation files and the graphical image data are Graphics Interchange Format (GIF) files. In an implementation, consistent with the invention, the slide **102** includes any known media in any other known media formats. In another implementation, the slide **100** includes numerical data, such as a table of numbers specified by the initial numerical data **430**. For example, the numerical data includes corporate profits over a range of years, with the initial text **434** including legends for the corporate profits and for the range of years. In a further implementation, the slide **100** includes a graph, a pie chart, or other images specified by the supplemental initial media data **436** which corresponds to and illustrates the initial numerical data **430**.

In an additional implementation consistent with the invention, the slide **100** is a webpage specified by the initial input data **404**, with the text **102** specified by the initial text **434** and with formatting specified by the format data **432**. The slide **100**, as a webpage, is generated using webpage authoring software which receives and processes the initial input data **404**. For example, the webpage stores the initial text **434** as the text **102** in a HyperText Markup Language (HTML). The webpage uses delimiters positioned on each side of a line or multiple lines of initial text **434**, such as the delimiters “< >” and “</>”. For example, the initial text **434** includes the tags such as “<body>” and “</body>” on either side of text **102** to be displayed on the webpage. In another example, the initial text **434** includes text corresponding to the text **102**, and the format data **432** in the initial input data **404** indicates the tags such as “<body>” and “</body>”. The webpage-based slide **100** is displayed by a web browser as a presentation software responsive to the initial input data **404**.

In an alternative implementation, the initial input data **404** is formatted in the Standard Generalized Markup Language (SGML), Extensible Markup Language (XML), or any known data format configured to be processed and then outputted or displayed by any known presentation software, such as POWERPOINT publicly available from MICROSOFT CORPORATION, as well as web browsers such as CHROME publicly available from GOOGLE LLC, or EDGE publicly available from MICROSOFT CORPORATION, and any other known data output device or software.

The input/output device **414** outputs the enhanced media presentation **406** generated by the presentation generating system **402**. In one implementation, the input/output device **414** includes a physical printer configured to generate and output a hardcopy of the enhanced media presentation **406**. For example, the enhanced media presentation **406** is an enhanced version of the slide **100** in FIG. 1, in which media as described above are added to the slide **100**. In one example, the enhanced media presentation **406** has text or an image added to the slide **100**. The hardcopy of the enhanced media presentation **406** has the added text or image included on the slide **100**. For example, the image added to the slide **100** and included in the enhanced media presentation **406** is a graphical image. The graphical image includes photos, graphs illustrating numerical values, uniform resource locator (URL) addresses of websites, quick-response (QR) codes encoding information such as URL addresses, barcodes, etc.

In another implementation, the input/output device **414** includes a display or monitor configured to generate and display the enhanced media presentation **406** with any visual media, including text, images, video, animation, or any known visual media added to the displayed slide **100**. In a

further implementation, the input/output device **414** includes a sound speaker configured to output any audio media, including sound, video, or any known audio media generated by the presentation generating system **402** to be associated with the slide **100** and so enhancing the slide **100**. In another implementation, the input/output device **414** is configured to output both visual media and audio media included in the enhanced media presentation **406** to enhance the slide **100**.

FIG. 5 illustrates a schematic of a computing device **500** including a processor **502** having code therein, a memory **504**, and a communication interface **506**. Optionally, the computing device **500** can include a user interface **508**, such as an input device, an output device, or an input/output device. The processor **502**, the memory **504**, the communication interface **506**, and the user interface **508** are operatively connected to each other via any known connections, such as a system bus, a network, etc. Any component, combination of components, and modules of the system **400** in FIG. 4 can be implemented by a respective computing device **500**. For example, the processor **408**, the memory **410**, the communication interface **412**, the input/output device **414**, and each of the set of modules **416-428** including the extraction module **416**, the centralization module **418**, the prompt generating module **420**, the presentation generating module **422**, the first generative AI module **424**, the second generative AI module **426**, and the third generative AI module **428** shown in FIG. 4 can be implemented by a respective computing device **500** shown in FIG. 5 and described below.

It is to be understood that the computing device **500** can include different components. Alternatively, the computing device **500** can include additional components. In another alternative implementation, some or all of the functions of a given component can instead be carried out by one or more different components. The computing device **500** can be implemented by a virtual computing device. Alternatively, the computing device **500** can be implemented by one or more computing resources in a cloud computing environment. Additionally, the computing device **500** can be implemented by a plurality of any known computing devices.

The processor **502** can be a hardware-based processor implementing a system, a sub-system, or a module. The processor **502** can include one or more general-purpose processors. Alternatively, the processor **502** can include one or more special-purpose processors. The processor **502** can be integrated in whole or in part with the memory **504**, the communication interface **506**, and the user interface **508**. In another alternative implementation, the processor **502** can be implemented by any known hardware-based processing device such as a controller, an integrated circuit, a microchip, a central processing unit (CPU), a microprocessor, a system on a chip (SoC), a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC). In addition, the processor **502** can include a plurality of processing elements configured to perform parallel processing. In a further alternative implementation, the processor **502** can include a plurality of nodes or artificial neurons configured as an artificial neural network. The processor **502** can be configured to implement any known machine learning (ML) based devices, any known artificial intelligence (AI) based devices, and any known artificial neural networks, including a convolutional neural network (CNN).

The memory **504** can be implemented as a non-transitory computer-readable storage medium such as a hard drive, a solid-state drive, an erasable programmable read-only

memory (EPROM), a universal serial bus (USB) storage device, a floppy disk, a compact disc read-only memory (CD-ROM) disk, a digital versatile disc (DVD), cloud-based storage, or any known non-volatile storage.

The code of the processor 502 can be stored in a memory internal to the processor 502. The code can be instructions implemented in hardware. Alternatively, the code can be instructions implemented in software. The instructions can be machine-language instructions executable by the processor 502 to cause the computing device 500 to perform the functions of the computing device 500 described herein. Alternatively, the instructions can include script instructions executable by a script interpreter configured to cause the processor 502 and computing device 500 to execute the instructions specified in the script instructions. In another alternative implementation, the instructions are executable by the processor 502 to cause the computing device 500 to execute an artificial neural network. The processor 502 can be implemented using hardware or software, such as the code. The processor 502 can implement a system, a subsystem, or a module, as described herein.

The memory 504 can store data in any known format, such as databases, data structures, data lakes, or network parameters of a neural network. The data can be stored in a table, a flat file, data in a filesystem, a heap file, a B+ tree, a hash table, or a hash bucket. The memory 504 can be implemented by any known memory, including random access memory (RAM), cache memory, register memory, or any other known memory device configured to store instructions or data for rapid access by the processor 502, including storage of instructions during execution.

The communication interface 506 can be any known device configured to perform the communication interface functions of the computing device 500 described herein. The communication interface 506 can implement wired communication between the computing device 500 and another entity. Alternatively, the communication interface 506 can implement wireless communication between the computing device 500 and another entity. The communication interface 506 can be implemented by an Ethernet, Wi-Fi, Bluetooth, or USB interface. The communication interface 506 can transmit and receive data over a network and to other devices using any known communication link or communication protocol.

The user interface 508 can be any known device configured to perform user input and output functions. The user interface 508 can be configured to receive an input from a user. Alternatively, the user interface 508 can be configured to output information to the user. The user interface 508 can be a computer monitor, a television, a loudspeaker, a computer speaker, or any other known device operatively connected to the computing device 500 and configured to output information to the user. A user input can be received through the user interface 508 implementing a keyboard, a mouse, or any other known device operatively connected to the computing device 500 to input information from the user. Alternatively, the user interface 508 can be implemented by any known touchscreen. The computing device 500 can include a server, a personal computer, a laptop, a smartphone, or a tablet.

Referring to FIGS. 4 and 6-7, in an implementation consistent with the invention, each of the first generative AI module 424, the second generative AI module 426, and the third generative AI module 428 are implemented using the large learning model (LLM) 600 shown in FIG. 6 having a neural network 602 utilizing a transformer architecture such as the transformer module 700 shown in FIG. 7. In one

implementation, the large learning model 600 is the GENERATIVE PRE-TRAINED TRANSFORMER 4 (GPT-4) publicly available from OPENAI, INC.

As shown in FIG. 6, the neural network 602 includes a plurality of nodes or artificial neurons 604 arranged in a plurality of layers 606, 608, 610, 612, 614. The layer 606 is an input layer, and the layer 614 is an output layer, with the layers 608, 610, 612 being at least one hidden layer between input layer 606 and the output layer 614. In an implementation consistent with the invention, the neural network 602 implementing the transformer module 700 shown in FIG. 7 is an N layer transformer model with a hidden layer size of H, in which N and H are integers greater than or equal to one. In one implementation, the values of N and H are predetermined values. For example, N equals 120 and H equals 12,288, and so the transformer model has 120 layers with a hidden layer size of 12,228. In another implementation, the values of N and H are set or changed by a system administrator by inputting desired values N and H using the input/output device 414 to configure the transformer model to have N overall layers, and to configure hidden layers of the transformer model to have H hidden layers. The values of N and H are stored in the memory 410.

In an implementation, as shown in FIG. 7, the transformer module 700 includes a tokenization module 706, a vector representation module 708, a first normalization module 710, a first multi-head attention module 712, a first feedforward and summation module 714, at least a second normalization module 716, at least a second multi-head attention module 718, at least a second feedforward and summation module 720, and an un-embedding layer 722. The tokenization module 706 generates tokens corresponding to the input data 702. The vector representation module 708 acts as an embedding layer, which converts the tokens and positions of the tokens into vector representations as vectorized chunks of the input data 702. In an implementation, the vector representations are stored in a vector embedding database in the memory 410 in FIG. 4. Multiple sets of the components 710-720 are chained to carry out repeated transformations on the vector representations, extracting more and more linguistic information, using alternating attention and feedforward layers. The final transformed vector representations are converted by the un-embedding layer 722 back to a probability distribution over the tokens to generate the transformed data 704.

Referring to FIGS. 4 and 8, in an implementation consistent with the invention, the system 400 operates such that the presentation generating system 402 receives the initial input data 404 through the communication interface 412, and saves the initial input data 404 in the memory 410. In another implementation, the processor 408 processes a slide 100 having the initial input data 404, and generates the initial input data 404. For example, the processor 408 uses the TYPESCRIPT application publicly available from MICROSOFT CORPORATION to generate the initial input data 404 from an input slide 100. As shown in FIG. 8, the extraction module 416 receives the initial input data 404, and uses the first generative AI module 424 to extract a plurality of features 800 from the initial input data 404. In an implementation, features 800 are in a text-based format.

In an implementation, the first generative AI module 424 includes a theme extraction module 802, an adjective extraction module 804, a color extraction module 806, and an image style extraction module 808. The theme extraction module 802 is configured to extract P themes from the initial input data 404. The adjective extraction module 804 is configured to extract Q adjectives from the initial input data

404. The color extraction module **806** is configured to extract R colors from the initial input data **404**. The image style extraction module **808** is configured to extract S image styles from the initial input data **404**. In an implementation, each of the parameters P, Q, R, and S are integers greater than or equal to one. For example, as shown in FIG. 9, the default value of P is one, while the default values of Q, R, and S are five. In an implementation, a system administrator uses the input/output device **414** to input, set, or change the parameters P, Q, R, and S stored in the memory **410**.

In another implementation, the first generative AI module **424** optionally includes a supplemental feature extraction module **810** configured to extract T supplemental features from the initial input data **404**. The parameters T is an integer greater than or equal to one. The default value of T is one. Such supplemental features include an image, a video, a sound, a sound style such as a type of music, three-dimensional (3D) video, 3D objects, QR codes encoding website information, URLs, barcodes, etc. as output formats. In an implementation, a system administrator uses the input/output device **414** to input, set, or change the parameter T stored in the memory **410**.

In one implementation, the extraction module **416** applies the initial input data **404** to the first generative AI module **424** to perform the extraction of the plurality of features **800** from the initial input data **404**. The plurality of features **800** include a plurality of terms. In an implementation, the terms are in a text-based format. As shown in FIG. 9, a data structure **900** includes a set of extracted features **902**, a default number **904** of terms to be extracted from the slide **100**, and extraction instructions **906** to be applied to the first generative AI module **424**. For example, the data structure **900** is a table. In another example, the data structure **900** is an array of records. In a further example, the data structure **900** is any known arrangement of data to be used by the first generative AI module **424**. In one implementation, the data structure **900** is stored in the memory **410**. In another implementation, the data structure **900** is stored in a memory of the extraction module **416**.

In an implementation, the extraction module **416** uses extractions instructions, such as the example extraction instructions **906**, shown in FIG. 9, to process the initial input data **404**. In one implementation, the extraction instructions **906** are text in a text-based format. For example, the extraction instructions **906** are stored in the memory **410**. In another example, the extraction instructions are stored in a memory of the extraction module **416**. In one implementation, the extraction instructions **906** are created by a system administrator or a technician using the input/output device **414**. In another implementation, the extraction instructions are pre-generated extraction instructions received from a third-party data source through the communication interface **412**. The extraction module **416** inputs the initial input data **404** to the first generative AI module **424** to generate the terms for each feature, such as the example terms **1004** for each feature as shown in FIG. 10, under the guidance, control, or constraints of the extraction instructions **906**. The extraction instructions **906** guide, control, or constrain the first generative AI module **424** of the extraction module **416** by focusing the extraction module **416** on the terms **1004** determined and defined by the various extraction modules **802**, **804**, **806**, **808**, **810** instead of having the first generative AI module **424** seeing and analyzing the whole input. Accordingly, the extraction instructions **906** create an artificial “keyhole” view of the content without explicit fine-

tuning for the kind of content. In an implementation, the plurality of features **1002** and terms **1004** are in a text-based format.

In particular, the initial input data **404** are the input data **702**, as text, applied to the transformer **700** of the first generative AI module **424** shown in FIG. 7. The first generative AI module **424** generates the extracted features **800** as the transformed data **704**, as text, output by the transformer **700** of the first generative AI module **424** shown in FIG. 7.

In one implementation, the extraction module **416** generates the plurality of extracted features **800** having a text format. For example, as shown in FIG. 10, a table **1000** includes a set **1002** of extracted features, and a set of terms **1004** describing each of the extracted features **800**, with the sets **1002**, **1004** having a text format. In one implementation, the set **1002** of extracted features includes themes, adjectives, color, and image styles, and the set **1004** of terms include “investment” as the extracted theme; “committing”, “additional”, “profitable”, “important”, and “investing” as the extracted adjectives; “green”, “gold”, “blue”, “grey”, and “white” as the extracted colors; and “financial”, “educational”, “infographic”, “professional”, and “simplicistic” as the extracted image styles. In another implementation, the extraction module **416** generates the plurality of extracted features **800** as text in the HyperText Markup Language (HTML), the Standard Generalized Markup Language (SGML), or the Extensible Markup Language (XML), with delimiters positioned on each side of the extracted features, such as the delimiters “<” and “>”. By using delimiters, the extracted features **800** are easily processed. In a further implementation, the extraction module **416** generates the plurality of extracted features **800** with each of the extracted features **800** associated with tags such as “<color>” and “</color>” on either side of an extracted color for ease of processing. In still another implementation, the extraction module **416** generates the plurality of extracted features **800** in any known data format.

The extracted plurality of features **800** are transmitted, conveyed, or otherwise provided to the centralization module **418**. In one implementation, the centralization module **418** collects such extracted features **800** in the memory **410**. In another implementation, the centralization module **418** collects such extracted features **800** in a memory of the centralization module **418**. The centralization module **418** transmits, conveys, or otherwise provides the extracted features **800** to the prompt generating module **420**. In another implementation, the centralization module **418** formats the features **800** into a predetermined format compatible for processing by the prompt generating module **420**.

In an implementation, the prompt generating module **420** uses prompt instructions, such as the set **1100** of prompt instructions, shown in FIG. 11, to process the extracted features **800**. In one implementation, the prompt instructions are text in a text-based format. For example, the prompt instructions are stored in the memory **410**. In another example, the prompt instructions are stored in a memory of the prompt generating module **420**. In one implementation, the prompt instructions are created by a system administrator or a prompt engineering technician using the input/output device **414**. In another implementation, the prompt instructions are pre-generated prompt instructions received from a third-party data source through the communication interface **412**. The prompt generating module **420** inputs the extracted features **800** shown in FIG. 8, to the second generative AI module **426** to generate a prompt, such as the example prompt **1200** as shown in FIG. 12, under the guidance,

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control, or constraints of the set **1100** of prompt instructions. In an implementation, the prompt **1200** is in a text-based format.

In particular, the extracted features **800** are the input data **702**, as text, applied to the transformer **700** of the second generative AI module **426** shown in FIG. 7. The second generative AI module **426** generates the prompt **1200** as the transformed data **704**, as text, output by the transformer **700** of the second generative AI module **426** shown in FIG. 7. The set **1100** of prompt instructions guide, control or constrain the second generative AI module **426** during prompt generation by enforcing rules on how the second generative AI module **426** is to use data from the table **1000**, such as the set **1002** of features, by explicitly maximizing the term count use per prompt.

As shown in FIG. 11, the set **1100** of prompt instructions include various aspects of how a prompt is to be created by the second generative AI module **426**. For example, the set **1100** of instructions includes a basic instruction “create a prompt for a scene using the extracted features and terms”. In another example, the granularity of the degree of detail is specified in the set **1100** of instructions, such as “The prompt is to describe every detail of a scene”. In a further example, the set **1100** of instructions specify a minimum number of extracted features to be used, such as “The prompt should use at least one theme, at least one adjective, at least one color, and at least one image among the terms provided.” In an additional example, the set **1100** of instructions provide an abstract degree of flexibility in creating the prompt, such as “The scene should be minimalistic and creative.” In still another example, the set **1100** of instructions impose at least one constraint on the created prompt, such as “The prompt should be at most 150 words in length.”

In one implementation, the prompt generating module **420** uses the Chain-of-Thought (CoT) prompting technique, the Chain-of-Symbol (CoS) prompting technique, or any known prompting technique. In another implementation, the prompt generating module **420** uses the Retrieval-Augmented Generation (RAG) technique, the Graph Retrieval-Augmented Generation (GraphRAG) technique, or any other known automatic prompt generation technique.

As shown in the example prompt **1200** in FIG. 12, the extracted features **800**, such as the example extracted features **1004**, are included in the generated prompt **1200**. For illustrative purposes, the example extracted features **1004** are in bold and italics. Once the prompt **1200** shown in FIG. 12 is generated from the extracted features **800** using the prompt instructions **1100**, the prompt **1200** is transmitted, conveyed, or otherwise provided to the presentation generating module **422** which includes the third generative AI module **428**. In one implementation, the third generative AI module **428** is a text-to-image generator, such as the DALL-E model, publicly available from OPENAI, which is a text-to-image model using deep learning methodologies to generate digital images from natural language descriptions such as the prompt **1200** shown in FIG. 12. In another implementation, the third generative AI module **428** is any known text-to-media generator, in which the media is text, images, audio, animation, video, or any other known media.

In an implementation consistent with the invention, the presentation generating module **422** includes a known presentation application or software such as POWERPOINT publicly available from MICROSOFT CORPORATION. In an implementation, the presentation application operates in conjunction with the third generative AI module **428**. In another implementation, the presentation application is integrated with the third generative AI module **428**.

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The presentation generating module **422**, using the third generative AI module **428** processing the prompt **1200**, generates enhancements to the slide **100** corresponding to the initial input data **404** having the associated extracted features **800**. In particular, the prompt **1200** is the input data **702**, as text, applied to the transformer **700** of the third generative AI module **428** shown in FIG. 7. The third generative AI module **428** generates the enhancements as the transformed data **704**, for example, text, images, audio, animation, video, or any other known media, output by the transformer **700** of the third generative AI module **428** shown in FIG. 7.

In an implementation consistent with the invention, in response to the prompt **1200**, the presentation generating module **422**, using the third generative AI module **428**, generates at least one image **1300**, **1302**, **1304**, as an enhancement media, shown in FIGS. 13A-13C, respectively, based on the prompt **1200** and the extracted features **800** therein. In another implementation, the third generative AI module **428** generates an enhancement media in any known format, including text, an image, audio, animation, video, numerical data, a graph representing numerical data, or any other known media. Depending on a value of a temperature parameter of the third generative AI module **428**, different images **1300**, **1302**, **1304** are generated. Temperature is a parameter that controls the creativity or randomness of the text generated by a generative AI system, such as the third generative AI module **428**. In an implementation, the temperature is in the range from 0.50 to 1.0, inclusive. In one implementation, the temperature is randomized, for example, using a known random number generator application. In another implementation, the temperature is set to a default value, such as 0.7 which is stored in the memory **410**. In a further implementation, a system administrator uses the input/output device **414** to input, set, or change the temperature value stored in the memory **410**. A higher temperature results in more diverse and creative output, while a lower temperature causes the output to be more deterministic and focused. The temperature affects the probability distribution over the possible tokens at each step of the generation process by the transformer module **700** shown in FIG. 7. A temperature of zero would make the third generative AI module **428** completely deterministic and always choosing the most likely token.

Depending on the temperature value, the third generative AI module **428** generates an image **1300**, **1302**, **1304** corresponding to the temperature value. In one implementation, the presentation generating module **422** formats the generated image **1300**, **1302**, **1304** into a suitable image format for use by the presentation application or software such as POWERPOINT publicly available from MICROSOFT CORPORATION. For example, a generated image **1300**, **1302**, **1304** is formatted to be in a Base64 encoded uniform resource identifier (URI) format for incorporation into the enhanced media presentation **406** displayed by POWERPOINT.

The presentation generating module **422** generates the enhanced media presentation **406**, being the slide **100** enhanced by media enhancements generated by the presentation generating system **402**. For example, using the image **1306** generated by the third generative AI module **428** as described above, the presentation generating module **422** generates the enhanced media presentation **1400** shown in FIG. 14 which is output by the presentation generating system **402** as the enhanced media presentation **406**. As shown in FIG. 14, the enhanced media presentation **1400**

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includes the text **1402** from the original slide **100** complemented or supplemented by the image **1404** derived from the image **1306**.

Referring to FIG. **15**, a computer-based method **1500** includes receiving the initial input data **404** representing a media presentation to be enhanced in step **1502**, extracting at least one term for at least one feature from the initial input data **404** using the first generative AI module **424** in step **1504**, collecting the extracted terms **1004** using the centralization module **418** in step **1506**, provide a prompt instruction **1100** in step **1508**, generating a prompt **1200** from the extracted terms **1004** and the prompt instruction **1100** using the second generative AI module **426** in step **1510**, generating an enhancement **1302**, **1304**, **1306** from the prompt **1200** using a third generative AI module **428** in step **1512**, and generating and outputting an enhanced media presentation **406** from the initial input data **404** and the enhancement **1302**, **1304**, **1306** in step **1514**.

In an implementation consistent with the invention, a non-transitory computer-readable storage medium stores instructions executable by a processor, such as the processor **408**, to generate the enhanced media presentation **406** from the initial input data **404**. The instructions include the steps **1502-1514** of the method **1500** shown in FIG. **15**.

Portions of the methods described herein can be performed by software or firmware in machine readable form on a tangible or non-transitory storage medium. For example, the software or firmware can be in the form of a computer program including computer program code adapted to cause the system to perform various actions described herein when the program is run on a computer or suitable hardware device, and where the computer program can be implemented on a computer readable medium. Examples of tangible storage media include computer storage devices having computer-readable media such as disks, thumb drives, flash memory, and the like, and do not include propagated signals. Propagated signals can be present in a tangible storage media. The software can be suitable for execution on a parallel processor or a serial processor such that various actions described herein can be carried out in any suitable order, or simultaneously.

It is to be further understood that like or similar numerals in the drawings represent like or similar elements through the several figures, and that not all components or steps described and illustrated with reference to the figures are required for all embodiments, implementations, or arrangements.

The terminology used herein is for the purpose of describing particular implementations only and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “contains,” “containing,” “includes,” “including,” “comprises,” and/or “comprising,” and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Terms of orientation are used herein merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In addition, the use of ordinal numbers (e.g., first, second, third) is for distinction and not counting. For example, the use of “third” does

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not imply there is a corresponding “first” or “second.” Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

While the disclosure has described several exemplary implementations, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to implementations of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular implementations disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all implementations falling within the scope of the appended claims.

The subject matter described above is provided by way of illustration only and should not be construed as limiting. Various modifications and changes can be made to the subject matter described herein without following the example embodiments, implementations, and applications illustrated and described, and without departing from the true spirit and scope of the invention encompassed by the present disclosure, which is defined by the set of recitations in the following claims and by structures and functions or steps which are equivalent to these recitations.

What is claimed is:

1. A presentation enhancement system configured to enhance an initial media presentation having initial text, comprising:

- a hardware-based processor;
- a memory configured to store instructions and configured to provide the instructions to the hardware-based processor; and
- a set of modules configured to implement the instructions provided to the hardware-based processor, the set of modules including:
 - an extraction module including a first generative artificial intelligence (AI) module configured to automatically extract a plurality of features from the initial text;
 - a prompt generating module including a second generative AI module configured to automatically generate a prompt from the plurality of features; and
 - a presentation generating module including a third generative AI module configured to automatically generate enhancement media from the prompt using a temperature parameter having a value set to a predetermined temperature value between 0.5 and 1.0, inclusive, and to automatically generate and output an enhanced media presentation combining the initial text with the enhancement media.

2. The presentation enhancement system of claim 1, wherein the third generative AI module automatically generates an image as the enhancement media from the prompt.

3. The presentation enhancement system of claim 1, wherein the first and second generative AI modules are text-to-text generative AI modules, wherein the plurality of features and the prompt are in a text format, and

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wherein the third generative AI module is a text-to-image generative AI module configured to automatically generate an image as the enhancement media from the text-formatted prompt.

4. The presentation enhancement system of claim 1, wherein each of the first, second, and third generative AI modules includes a neural network configured as a transformer to implement a large language model.

5. The presentation enhancement system of claim 1, wherein the plurality of features includes a theme, an adjective, a color, and an image style.

6. The presentation enhancement system of claim 5, wherein the extraction module further comprises:

a theme extraction module configured to extract the theme from the initial text;

an adjective extraction module configured to extract the adjective from the initial text;

a color extraction module configured to extract the color from the initial text; and

an image style extraction module configured to extract the image style from the initial text.

7. The presentation enhancement system of claim 1, wherein the value of the predetermined temperature parameter is set to a randomized temperature value between 0.5 and 1.0, inclusive.

8. A presentation enhancement system configured to enhance an initial media presentation having initial text, comprising:

a hardware-based processor;

a memory configured to store instructions and configured to provide the instructions to the hardware-based processor; and

a set of modules configured to implement the instructions provided to the hardware-based processor, the set of modules including:

an extraction module including a first generative artificial intelligence (AI) module implementing a first large language model and configured to automatically extract a plurality of features from the initial text;

a prompt generating module including a second generative AI module implementing a second large language model and configured to automatically generate a prompt from the plurality of features; and

a presentation generating module including a third generative AI module implementing a third large language model and configured to automatically generate enhancement media from the prompt using a temperature parameter having a value set to a predetermined temperature value between 0.5 and 1.0, inclusive, and to automatically generate and output an enhanced media presentation combining the initial text with the enhancement media.

9. The presentation enhancement system of claim 8, wherein the third generative AI module automatically generates an image as the enhancement media from the prompt.

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10. The presentation enhancement system of claim 8, wherein the first and second generative AI modules are text-to-text generative AI modules,

wherein the plurality of features and the prompt are in a text format, and

wherein the third generative AI module is a text-to-image generative AI module configured to automatically generate an image as the enhancement media from the text-formatted prompt.

11. The presentation enhancement system of claim 8, wherein each of the first, second, and third generative AI modules includes a neural network configured as a transformer to implement the first, second, and third large language models, respectively.

12. The presentation enhancement system of claim 8, wherein the plurality of features includes a theme, an adjective, a color, and an image style.

13. The presentation enhancement system of claim 12, wherein the extraction module further comprises:

a theme extraction module configured to extract the theme from the initial text;

an adjective extraction module configured to extract the adjective from the initial text;

a color extraction module configured to extract the color from the initial text; and

an image style extraction module configured to extract the image style from the initial text.

14. The presentation enhancement system of claim 8, wherein the value of the predetermined temperature parameter is set to a randomized temperature value between 0.5 and 1.0, inclusive.

15. A computer-based method, comprising:

receiving initial input data representing an initial media presentation;

automatically extracting at least one term for at least one feature from the initial input data using a first generative artificial intelligence (AI) module;

providing a prompt instruction;

automatically generating a prompt from the extracted at least one term and from the prompt instruction using a second generative AI module;

automatically generating an enhancement media from the prompt using a third generative AI module and a temperature parameter having a value set to a predetermined temperature value between 0.5 and 1.0, inclusive; and

generating and outputting an enhanced media presentation from the initial input data and the enhancement media.

16. The computer-based method of claim 15, wherein each of the first, second, and third generative AI modules includes a neural network configured as a transformer to implement a large language model.

17. The computer-based method of claim 15, wherein the value of the predetermined temperature parameter is set to a randomized temperature value between 0.5 and 1.0, inclusive.

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